

Ablanvi Songo

Research Interests

My research is in the field of nonlinear analysis. I am particularly interested in linking theorems (for single-valued and multivalued functionals) in critical point theory, as well as their applications to the study of solutions to PDEs and partial differential inclusions using variational and topological methods. I am also interested in combining the theoretical analysis of ODEs and PDEs with their numerical solution.

Education

- Winter 2021 - **Ph.D. in Mathematics**, *Université de Sherbrooke, Canada*,
 Summer 2026 Title : Théorèmes de points critiques pour des fonctionnelles éventuellement non différentiables,
 Supervisors: Fabrice Colin, Vasilisa Shramchenko
 Grade: "Excellent"
- 2019-2020 **M.Sc. (1st year) in Actuarial and Financial Mathematics**, *Université de Rouen Normandie, France*
 2018-2019 **M.Sc. (2nd year) in Fundamental and Applied Mathematics**, *Université de Rouen Normandie, France*
 2017-2018 **M.Sc. (1st year) in Fundamental Mathematics**, *Aix Marseille Université, France*
 2011-2015 **B.Sc. in Mathematics**, *Université de Lomé, Togo*

Publications and Preprints

Published journal articles

- 2026 **Existence result for a system of two semilinear coupled Poisson equations with asymptotically linear nonlinearities**, *Differential Equations & Applications*, in Press
 2026 **An infinite dimensional Saddle Point Theorem and application**, with F. Colin, *Boundary Value Problems*, (2026), <https://doi.org/10.1186/s13661-026-02234-8>

Submitted journal articles

- 2026 **An infinite-dimensional Brouwer's fixed-point theorem**
 2026 **An infinite-dimensional mountain pass theorem with applications to nonlinear elliptic systems**, with F. Colin, <https://doi.org/10.48550/arXiv.2512.02229>
 2026 **On the Fountain Theorem for Continuous Functionals and Its Application to a Semilinear Elliptic Problem in R^2** , with F. Colin, <https://doi.org/10.48550/arXiv.2509.16059>
 2026 **A new infinite-dimensional Linking theorem with application to a system of coupled Poisson equations**, with F. Colin, <https://doi.org/10.48550/arXiv.2506.17563>

Preprints

- 2026 **Linking theorems for multivalued functionals and application to partial differential inclusions**, with F. Colin, <https://doi.org/10.48550/arXiv.2607.22506>
 2026+ **Mountain Pass Theorem for multivalued functionals and application to subcritical elliptic differential inclusion systems in R^2** , in preparation

Other Scientific achievements

- 2019 **Master's (M.Sc.) thesis**, *Université de Rouen Normandie, France*,
 Title: Les équations d'Hamilton et Jacobi sur les réseaux : Sous et Sur-solutions à flux-limité,
 Grade : "Très-bien"

Talks

- May 2026 **XXVIIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *UTQR*, Trois-Rivières
April 2026 **Présentations "180 secondes"**, *CaféFak*, *Faculté des sciences*, *Université de Sherbrooke*
November 2025 **Séminaire Junior en Géométrie et Physique Mathématique**, *Département de mathématiques*, *Université de Sherbrooke*
May 2025 **XXVIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *McGill University*
February 2025 **Club Math**, *Department of Mathematics*, *Université de Sherbrooke*
Nov 2023 **Séminaire "Séminaire" d'étudiants gradués**, *Department of Mathematics*, *Université de Sherbrooke*

Awards and Grants

- 2021-2024 **Doctoral scholarship of the Faculty of Sciences**, *Faculté des sciences*, *Université de Sherbrooke*
2021-2026 **ISM Travel Grant**, *Institut des Sciences Mathématiques (ISM)*, *Montréal*
Hiver 2026 **Award for Outstanding First Publication**, *Faculté des sciences*, *Université de Sherbrooke*
2023-2024, 2022-2023 **ISM Graduate Scholarship**, *Institut des Sciences Mathématiques (ISM)*, *Montréal*
2022-2023 **Bourse In memorian Ernest-Monga for perseverance and commitment**, *Département of Mathematics*, *Université de Sherbrooke*
2018-2019, 2017-2018 **Scholarship of Government of Togo**, *Ministère de l'Enseignement Supérieur et de la Recherche*

Conferences and Schools Attended

- May 2026 **XXVIIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *UTQR*, Trois-Rivières
November 2025 **Séminaire Junior en Géométrie et Physique Mathématique**, *Département de mathématiques*, *Université de Sherbrooke*
October 2025 **3rd University of North Carolina at Greensboro Virtual PDE Conference : dedication to Professor Alfonso Castro in celebration of his 75th birthday and his outstanding contributions to Differential Equations**, *Département of Mathematics and Statistics (online)*
May 2025 **XXVIIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *McGill University*
June 2024 **XXVIe colloque panquébécois de l'Institut des Sciences Mathématiques**, *Université de Québec à Montréal*
October 2023 **Black Heroes of Mathematics Conference 2023**, *ICMS*, *the University of Edinburgh (online)*
June 2023 **XXVe colloque panquébécois de l'Institut des Sciences Mathématiques**, *Université de Sherbrooke*
July 2022 **Topological Methods in Nonlinear Analysis: Recent Advances - Conference in memory of Professor Andrzej Granas**, *Centre de recherches mathématiques (CRM)*, *Université de Montréal*
May 2022 **XXIVe colloque panquébécois de l'Institut des Sciences Mathématiques**, *Université Laval*

Teaching Experience

Université de Sherbrooke

- Winter 2026 **Teaching Assistant**, *MAT 117 (Differential and Integral Calculus of Several Variables and Vector Analysis)*
Fall 2025 **Teaching Assistant**, *MAT 098 (Supplementary Material on Integral Calculus)*
Summer 2025 **Teaching Assistant**, *MAT 455 (Analysis III - Differential and Integral Calculus in $\mathbb{R}^n$)*
Winter 2025 **Teaching Assistant**, *MAT 129 (Analysis I - Elements of Analysis)*
Fall 2024 **Teaching Assistant**, *MAT 199 (Linear Algebra Applied to Computer Science)*, *MAT 221*, *CQP 208*
Winter 2024 **Teaching Assistant**, *MAT 129*, *MAT 221 (Differential and Integral Calculus)*, *CQP 208*
Fall 2023 **Teaching Assistant**, *MAT 221 (Differential and Integral Calculus)*
Winter 2023 **Teaching Assistant**, *MAT 129 (Elements of Analysis)*
Fall 2022 **Teaching Assistant**, *MAT 221 (Differential and Integral Calculus)*, *MAT 324*
Summer 2022 **Teaching Assistant**, *MAT 455 (Analysis III)*
Winter 2022 **Teaching Assistant**, *MAT 193 (Linear Algebra)*
Fall 2021 **Teaching Assistant**, *MAT 324 (Mathematical Models)*

Winter 2021 **Teaching Assistant, MAT 193 (Linear Algebra)**
Lycée de Davié (Togo)

2014-2017 **Nationally certified teacher, Mathematics, Physics-Chemistry**

Academic and Community Involvement

March 2026 **Volunteer during the presentation of mathematical games at the Sherbrooke Museum of Nature and Science organized by our Department, Sherbrooke**

May 2025 **Volunteer during the final round of the International Mathematical and Logical Games Championship (province of Quebec), Cegep de Sherbrooke**

Mars 2025 **Volunteer during the semi-final round of the International Mathematical and Logical Games Championship (Sherbrooke Region), Department of Mathematics, Université de Sherbrooke**

March 2025 **Volunteer during the presentation of mathematical games at the Sherbrooke Museum of Nature and Science organized by our Department, Sherbrooke**

March 2024 **Volunteer during the semi-final round of the International Mathematical and Logical Games Championship (Sherbrooke Region), Department of Mathematics, Université de Sherbrooke**

March 2024 **Volunteer during the presentation of mathematical games at the Sherbrooke Museum of Nature and Science organized by our Department, Sherbrooke**

May 2023 **Volunteer during the Quebec final of the International Mathematical and Logical Games Championship, Department of Mathematics, Université de Sherbrooke**

Winter 2023 **Volunteer during the Faculty of Science Open House, Université de Sherbrooke**

Fall 2022 **Volunteer during the Faculty of Science Open House, Université de Sherbrooke**

2021-2023 **Volunteer mathematics tutor for Le Diplôme Avant la Médaille (DAM) academic support program for student-athletes**

Technical Strengths and Skills

Programming

Python, SQL, $\text{\LaTeX}$

Languages

French, English, Ewè

Academic References

Fabrice Colin, *Associate Professor*, Laurentian University

Email: fcolin@laurentian.ca

Vasilisa Shramchenko, *Full Professor*, Université de Sherbrooke

Email: Vasilisa.Shramchenko@USherbrooke.ca